

TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits

Francis M. Peña, Jett M. Lumabi, Alex A. Alferez, Referendo D. Soriano, Ph.D.

Department of Electronics and Computer Engineering

Ateneo de Naga University

Naga City, Philippines

Abstract—Residential electrical-fire prevention extends beyond clearing hard short circuits. Progressive hazards such as sustained overload, degraded plug-socket contact heating, and load-limited series arcing may develop locally before conventional upstream protection directly observes the fault mechanism. This paper presents a microcontroller-based complementary socket-level protection architecture that separates the evidence required for these hazards. The implementation integrates calibrated voltage, current, and temperature sensing on an ESP32-S3; deterministic persistence logic for overload; compensated socket-temperature estimation for degraded contact heating; and load-aware Random Forest TinyML inference for series arc-fault detection. Relay isolation, fault latching, local audio-visual alerts, and Firebase/PWA monitoring and labeling support were implemented, while relay protection remained local to the firmware. Validation produced mean absolute errors of 3.910V, 0.178A, and 1.02°C for voltage, current, and thermistor channels, respectively. The highest-stress overload case averaged 17.30A and opened the relay after 61.3s. Degraded-socket trials heated faster than normal-socket trials and reached reference terminal temperatures of 55–56°C in severe cases before or during relay action. Across 200 balanced normal and arc trials, the embedded arc model with firmware persistence achieved 97.50% accuracy, 98.97% precision, 96.00% recall, and 97.46% F1-score; held-out event-level arc reporting showed 84.0% event recall with one false-alarm event. These results support controlled laboratory feasibility for a complementary embedded protection pipeline, with certification-oriented testing and listed-device comparison left for future work.

Keywords—TinyML, smart plug, series arc fault detection, socket heating, overload protection, residential electrical safety.

I. INTRODUCTION

Electrical faults remain a persistent residential fire concern. Philippine fire incident reporting identifies electrical issues as a leading cause of fires, while broader home-fire statistics also associate electrical failure or malfunction with structure fires, injuries, and property losses [1], [2]. The engineering problem includes the progressive region between apparently normal operation and a developed fire-inducing fault, in addition to catastrophic short circuits.

Conventional residential protection remains essential. Branch-circuit breakers, leakage protection, surge protection, and certified wiring practice address fault classes for which they are designed. However, ordinary overcur-

rent protection is coordinated mainly around conductor thermal protection and high-current events; it lacks direct visibility of outlet contact temperature and low-current series-arcing evidence [3]–[5]. At a socket, a poor plug-contact interface can develop localized heating while the total load current remains within the ordinary operating range. A load-limited series arc can also disturb the waveform without producing a large RMS-current increase.

This study distinguishes three fire-inducing mechanisms. Sustained overload is a current-persistence and time-stress problem. Degraded plug-socket contact heating is a local thermal problem, where the hottest contact region may be hidden from an external thermistor. Series arcing is a stochastic, load-dependent waveform problem, because normal nonlinear loads can share some arc-like characteristics. Combining these mechanisms under one generic abnormal state obscures the relevant evidence: an RMS current threshold alone cannot represent localized heating or load-limited arcing.

The objective of this work was to design and evaluate an ESP32-S3-based socket-level embedded protection prototype for selected progressive residential fault conditions. Specifically, the work aimed to integrate calibrated voltage, current, and temperature sensing; implement deterministic overload and socket-heating protection; deploy load-aware TinyML series arc inference; provide local relay isolation, fault latching, and alerts; and support monitoring, event logging, and data labeling through Firebase and a Progressive Web App (PWA).

This paper contributes a hybrid embedded protection architecture that separates evidence by fault mechanism: current persistence for overload, compensated socket-temperature estimation for degraded-contact heating, and load-context-conditioned waveform-feature inference for series arcs. The evaluation is framed as a controlled laboratory feasibility study in electronics and computer engineering (ECE), with socket-level protection treated as complementary to breakers, arc-fault circuit interrupters (AFCIs), ground-fault circuit interrupters (GFCIs), wiring inspection, and code-compliant residential protection.

TABLE I
Fault Mechanisms and Evidence Used by the Implementation

Fault class	Fire-relevant mechanism	Evidence used for protection
Sustained overload	Thermal stress from persistent load current	RMS-current threshold plus approximately 60s persistence score
Socket heating	Localized poor-contact heating at plug-socket interface	Thermistor input plus compensated estimated socket temperature
Series arc	Load-limited intermittent conduction and waveform disturbance	Load context, waveform features, Random Forest arc inference, and persistence counter

II. DESIGN BASIS AND SYSTEM OVERVIEW

A. Target Fault Classes

The system was designed around progressive socket-level hazards. Overload was treated as a sustained load-current condition. A warning may be useful when the current is high, but immediate relay operation for every short transient or inrush event is inconsistent with time-current protection behavior. Thus, overload requires current persistence.

Degraded socket-contact heating was treated separately because poor contact pressure, corrosion, looseness, or oxidation can increase local contact resistance. Glowing-contact and receptacle-fire studies show that this kind of poor connection can persist while the circuit still carries load current [6], [7]. Recent poor-contact fire studies likewise show the role of contact pressure, localized heating, pyrolysis, and ignition progression [8], [9]. Consequently, a local thermal estimate is needed even when current alone cannot resolve the socket-contact condition.

Series arc faults were treated as a waveform-inference problem. A series arc is load-limited, so it may not create a breaker-level current rise. Instead, it can produce low-current gaps, current fluctuation, zero-region behavior, harmonic disturbance, and high-frequency content [5], [16]–[19]. Because normal loads can also be nonlinear, the arc decision benefits from load context and family-specific waveform interpretation [20], [21].

The design basis is summarized in Table I.

Each protection path asks a different question: whether current is persistently high, whether the socket interface is thermally abnormal, or whether the current waveform is repeatedly consistent with arcing under the present load context.

B. Hybrid Architecture

Fig. 1 shows the system architecture used in the prototype. The sensing layer measures line voltage, load current, and temperature near the plug/socket interface. The firmware layer performs calibration, filtering, RMS calculations, temperature estimation, waveform-feature



Fig. 1. System architecture of the TinyML Smart Plug.

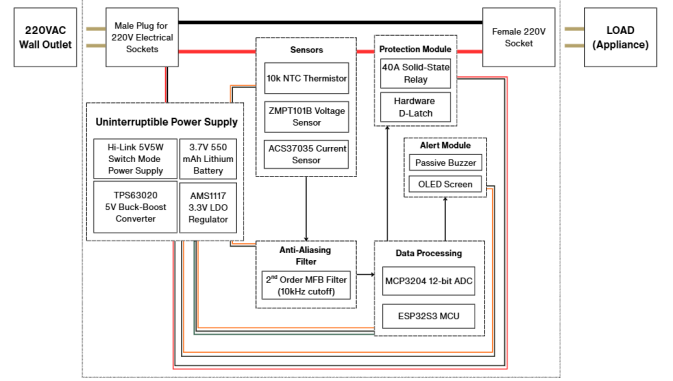


Fig. 2. Block diagram of the prototype hardware.

extraction, load-context classification, TinyML arc inference, protection-state evaluation, relay control, and event logging. The cloud/PWA layer provides visualization, history review, labeling, and operator interaction. The relay decision remains local to the firmware.

The architecture is intentionally hybrid. Deterministic thresholds are used where the protection evidence is direct and interpretable, as in overload current and estimated socket temperature. TinyML is used where the evidence is multi-feature, load-dependent, and ambiguous, as in series arc-fault detection. The resulting protection pipeline is local: if Wi-Fi or Firebase is unavailable, the ESP32-S3 can still evaluate faults and command relay isolation.

III. METHODOLOGY

A. Hardware and Sensing Architecture

Fig. 2 shows the block-level hardware implementation. The prototype was built as a plug-through socket-level device for a 230V residential circuit. Its major elements were an ESP32-S3 microcontroller, ZMPT101B voltage-sensing channel, ACS37035 current sensor, MCP3204 external 12-bit ADC, second-order anti-aliasing circuit, 10kΩ NTC thermistor, solid-state relay, relay latch, OLED display, buzzer, low-voltage supply, and enclosed outlet assembly.

The current channel was central to the design. It supplied RMS current for overload protection, a current-dependent term for socket-temperature estimation, and waveform samples for feature extraction. The conditioned current waveform was digitized through the MCP3204 and processed by the ESP32-S3. The voltage channel supported RMS-voltage supervision and mains-present interpretation. The thermistor was placed near the plug-side wire-contact region and insulated to reduce direct ambient-air influence. Because this point is offset from the hidden contact hotspot, the thermal protection was based

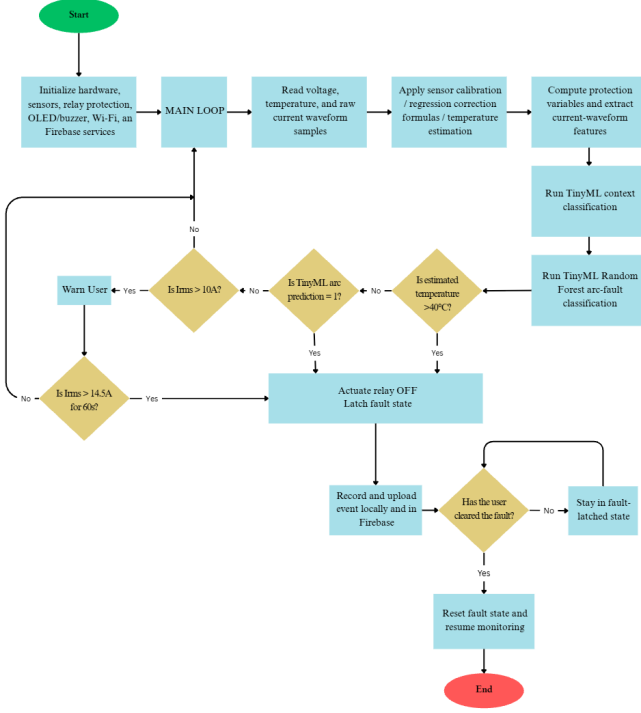


Fig. 3. Firmware protection flow from sensing and feature extraction to relay actuation and fault clearing.

on compensated estimated socket temperature, which represents the available external thermal measurement rather than a direct internal-contact measurement.

The relay and latch formed the final local protective actuator. After a confirmed trip condition, the relay disconnected the protected outlet and the fault state remained latched until cleared. The OLED, buzzer, and PWA provided alerts and user visibility. Firebase carried live telemetry, history, CSV review, and operator fields, but the relay actuation path was implemented in firmware and hardware.

B. Firmware Protection Flow

Fig. 3 summarizes the firmware protection flow implemented in the prototype. On startup, the ESP32-S3 initializes the sensors, ADC, relay driver, latch state, OLED, buzzer, Wi-Fi/Firebase interface, and protection manager. The main loop then acquires slow supervision variables and waveform frames in parallel: RMS voltage, RMS current, thermistor temperature, and current-frame features.

The final relay decision occurs in the protection manager. Calibrated voltage and current values feed deterministic voltage and overload states. The temperature routine produces device temperature and estimated socket temperature for thermal warning and trip evaluation. The waveform path computes feature frames, updates load context, runs Random Forest arc inference, and passes

TABLE II
Principal Firmware Protection Constants

Function	Warning or model condition	Relay-trip condition
Voltage supervision	Outside 200–250V band	Instant 170/260V, or 7s delayed low/high voltage
Overload	$I_{\text{rms}} > 10.0\text{A}$ warning	1s average current $\geq 14.5\text{A}$ for about 60s
Socket heating	39.0°C estimated temperature	Estimated socket temperature $\geq 40.0^\circ\text{C}$
Series arc fault	Context-aware Random Forest output	Arc counter reaches 8; +2 for arc frame, -6 for normal frame

the arc output to a leaky persistence counter. If overload, socket heating, or series arc evidence reaches its trip condition, the firmware pulses the relay latch, records the fault, and reports the fault state locally and through Firebase when available. After relay isolation, the plug remains in a latched fault state until the user clears the fault through local or PWA-supported interaction.

The principal protection constants are listed in Table II.

Socket heating had highest reported-fault priority because temperature is the most direct local fire-risk indicator. Arc fault was evaluated next because a series arc may be hazardous without sustained overcurrent. Overload remained a relay-trip condition after the persistence criterion was satisfied.

C. Sensor Calibration and Validation

Voltage and current calibration used polynomial correction because the raw sensor outputs showed offset and scale nonlinearity across the tested range. The voltage channel was calibrated and revalidated against a Fluke 115 True RMS multimeter. The current channel was calibrated using appliance load combinations and revalidated against an Ingco clamp meter. The thermistor channel used the NTC divider and Beta-equation conversion, then was compared against an infrared thermometer at the external thermistor measurement point.

The validation procedure assessed protection-grade suitability rather than traceable metrology. Voltage validation checked whether calibrated readings were adequate for supply-state supervision. Current validation checked whether RMS-current error was small relative to warning and trip thresholds. Temperature validation checked the thermistor measurement at its own location only. Hidden-contact temperature was treated separately through degraded-contact experiments and conservative thermal estimation, because standards and thermal-measurement literature caution that external temperatures may lag or under-read internal contact points [10]–[14].

D. Overload Protection Method

Overload testing used high-power load combinations connected through the prototype. The firmware issued warning-level alerts above 10.0A while retaining relay

closure during warning-level excursions. A sustained trip required the 1s current average to remain at or above 14.5A long enough for the overload score to reach six. Since the score increments every 10s, this corresponds to approximately 60s of persistent overload. This design choice prevents momentary current peaks and short transients from being treated as confirmed overload trips while still responding to persistent current stress.

The overload score can be represented by

$$S_{OL}[k] = \begin{cases} \min(6, S_{OL}[k-1] + 1), & I_{avg} \geq 14.5A, \\ \max(0, S_{OL}[k-1] - 1), & I_{avg} \leq 14.1A. \end{cases} \quad (1)$$

The relay trips when $S_{OL} = 6$. The method was evaluated by recording mean voltage, mean current, peak current, warning state, relay state, and trip latency during each load case.

E. Socket-Heating Method

Socket-heating evaluation used normal and intentionally degraded socket contacts under matched load cases. The degraded-contact trials served as accelerated laboratory comparisons rather than direct household field-fault replicas, allowing controlled observation of contact-heating behavior. For each trial, the prototype recorded device temperature from the external thermistor, estimated socket temperature from firmware, and relay state. An infrared thermometer supplied a reference terminal or contact-region reading for interpretation.

The firmware thermal model separated three thermal quantities. First, T_{NTC} was the direct external thermistor or device temperature. Second, $T_{normal,exp}$ was the expected normal socket temperature for a healthy contact carrying the measured current history. Third, $T_{socket,est}$ was the relay-facing estimated socket temperature used by the protection manager. This separation is needed because the thermistor is outside the hidden contact hotspot. It may lag or under-read the terminal region, while poor contact can generate localized heating that RMS current alone cannot represent.

The estimator was implemented as a compact firmware model. The current-shape term was computed as

$$s_I = \left[\text{clamp} \left(\frac{I_{rms}}{12.0}, 0, 1.25 \right) \right]^2, \quad (2)$$

so the thermal term follows the current-squared tendency of resistive heating while limiting extreme extrapolation. The expected normal term used a slowly filtered healthy-contact rise, whose active-load target was $\Delta T_{normal,target} = 0.20 + 12.0s_I$, added to the tracked ambient or baseline temperature T_a :

$$T_{normal,exp} = \text{clamp}(T_a + \Delta T_{normal}, -20, 125). \quad (3)$$

For an active load, the measured-temperature compensation was

$$\Delta T_{meas} = 0.30 + 1.20s_I, \quad (4)$$

with the current-dependent rise forced toward zero when the load was inactive. The protection-facing estimate was

$$\begin{aligned} T_{comp} &= \text{clamp}(T_{NTC} + \Delta T_{meas}, -20, 125), \\ T_{socket,est} &= \text{clamp}(\max(T_{comp}, T_{normal,exp}), -20, 125). \end{aligned} \quad (5)$$

The $\max(\cdot)$ operation prevents the relay-facing estimate from falling below the expected normal socket heating for the same load while still allowing measured local heating to dominate. The trip rule was applied to $T_{socket,est}$ so that raw thermistor readings and current-dependent normal heating both influenced the thermal state. The estimate should be interpreted as a conservative relay-facing variable, with its behavior checked against normal and corroded-socket experiments under comparable loads.

F. TinyML and Arc-Fault Method

Series arc detection used a two-stage load-aware TinyML pipeline. The method followed the firmware order: current-frame acquisition, waveform feature extraction, load-context classification, arc inference, and persistence-filtered relay evaluation. The firmware computed 512-sample current frames at a 27kHz target sampling rate with a 256-sample hop. A Hann-windowed frequency analysis and time-domain calculations generated waveform features; the Hann window was used to reduce spectral leakage in discrete Fourier analysis [25]. Logged features supported training, PWA review, and manual labeling.

The resulting TinyML processing sequence is shown in Fig. 4.

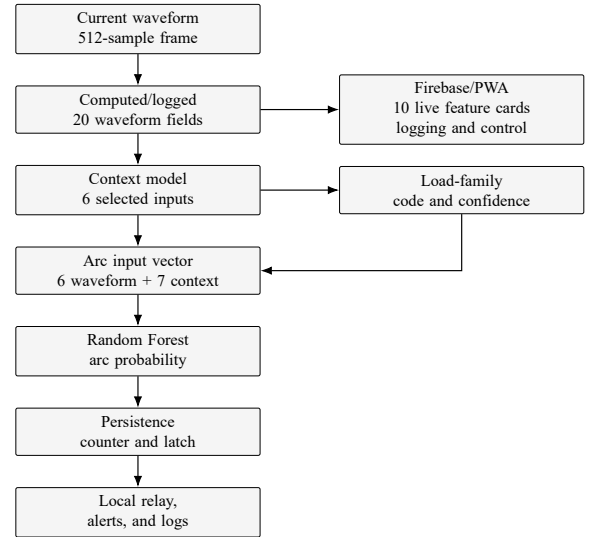


Fig. 4. TinyML processing sequence from waveform features to local relay action.

In this sequence, a centroid-based context model classified load family using six features: absolute RMS-current change, half-cycle asymmetry, midband residual ratio, current THD, mid/high-frequency spectral flux,

and high-frequency energy delta. The firmware context schema defined six possible family fields: resistive linear, inductive motor, rectifier SMPS, phase-angle controlled, brush/universal motor, and other mixed. However, the collected and tested signatures in this study were limited to resistive loads, inductive loads, and brushed or universal-motor loads. The context was accepted only when confidence was at least 0.45.

The arc model was a Random Forest deployed as a C header in firmware. Random Forest was selected because it can represent nonlinear feature interactions while remaining exportable as a compact embedded ensemble [23], [24]. The deployed model used 13 inputs: six waveform features and seven context fields. The waveform features were maximum low-current run, zero-dwell ratio, low-current ratio, absolute RMS-current z-score from baseline, midband residual ratio, and mid/high-frequency spectral flux. The context fields were six one-hot load-family inputs and one context-confidence input. The arc probability threshold was 0.48 for supported confident contexts and 0.78 for unknown context, with an additional minimum-feature-vote safeguard for unknown loads. The deployed feature set can be summarized in four groups. First, the conduction-gap inputs—maximum low-current run, zero-dwell ratio, and low-current ratio—represent extinction, weak conduction, and near-zero dwell that can occur during separation and restrike. Second, the current-departure input, expressed as an absolute RMS-current z-score from the recent baseline, measures load-relative departure from healthy operation. Third, the waveform-disturbance inputs, namely the midband residual ratio and mid/high-frequency spectral flux, represent residual spectral energy and frame-to-frame spectral change associated with nonstationary arcing. Finally, the load-context inputs, composed of six one-hot load-family fields and the context confidence, condition the arc decision so that normal nonlinear loads are evaluated with context-sensitive decision boundaries.

Training and testing were performed offline from firmware CSV logs. Records were reviewed, manually labeled, cleaned with Python scripts, split into training, validation, and held-out test views, and exported as firmware headers. The final arc report used 35,512 training rows, 9,434 validation rows, and 10,268 held-out test rows. Embedded testing then evaluated the exported model inside the firmware protection path, where the model output was filtered by the arc persistence counter before relay action. Thus, the reported trial result represents the deployed protection pipeline, including embedded inference and counter-based relay logic.

G. PWA, Firebase, and Logging Role

The PWA and Firebase backend supported the research workflow by showing live voltage, current, device temperature, estimated socket temperature, context family, arc output, relay status, event history, and waveform features.

They also supported CSV inspection, data labeling, and operator fault clearing. These functions improved observability and reproducibility during laboratory trials, but they were supporting infrastructure. The local ESP32-S3 firmware remained responsible for protection-state evaluation and relay isolation.

IV. RESULTS AND DISCUSSION

A. Sensor Validation

Table III summarizes the revalidation results. Voltage MAE was 3.910V, current MAE was 0.178A, and temperature MAE was 1.02°C. These errors were small relative to the implemented thresholds. Current error, for example, was much smaller than the separation between the 10.0A warning threshold and 14.5A sustained-trip threshold. The temperature result validates the external thermistor measurement point only; the socket-heating protection result is interpreted through the compensated estimate and degraded-contact tests.

TABLE III
Sensor Validation Summary

Channel	MAE	MSE	RMSE
Voltage	3.910V	15.539 V ²	3.94V
Current	0.178A	0.04076 A ²	0.202A
Temperature	1.02°C	1.142 °C ²	1.07°C

B. Overload Protection

Table IV gives the overload evaluation. All heavy-load trials exceeded the warning threshold, while relay operation occurred in the highest-stress case. Trial 4 averaged 17.30A, reached a 19.73A peak, and opened the relay after 61.3s. That trip time agrees with the intended approximately 60s persistence behavior. Trials 1, 3, and 5 reached peaks near or above the sustained threshold, then fell below the criterion before accumulating the required persistence score. Thus, the overload function behaved as time-aware complementary protection. Breaker coordination and formal time-current compliance testing require a separate standards-oriented evaluation.

TABLE IV
Overload Evaluation Results

Trial	Load setup	Mean A	Peak A	Relay trip
1	1.3kW kettle + water heater	12.87	14.85	No
2	1.3kW kettle + 1.5kW kettle	12.33	13.63	No
3	1.5kW kettle + water heater	13.36	14.86	No
4	1.3kW kettle + water heater + 2 lamps	17.30	19.73	61.3s
5	1.3kW kettle + water heater + 1 lamp	14.42	16.23	No

C. Socket-Heating Protection

Socket-heating characterization showed that contact condition changed the thermal outcome at similar current levels. The low-current protection case is shown in Fig. 5,

while the severe high-current case is shown in Fig. 6. The paired figures show the intended relay behavior across the tested range: lower-current degraded-contact heating remained below the trip criterion, while the severe case reached the relay-facing thermal threshold.

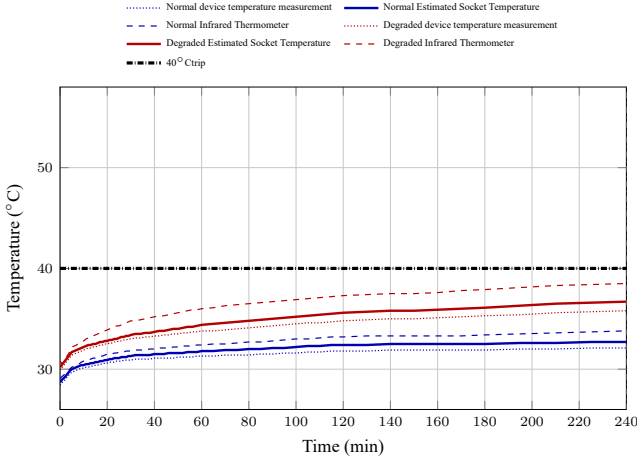


Fig. 5. Water-heater-only low-current socket-heating protection validation.

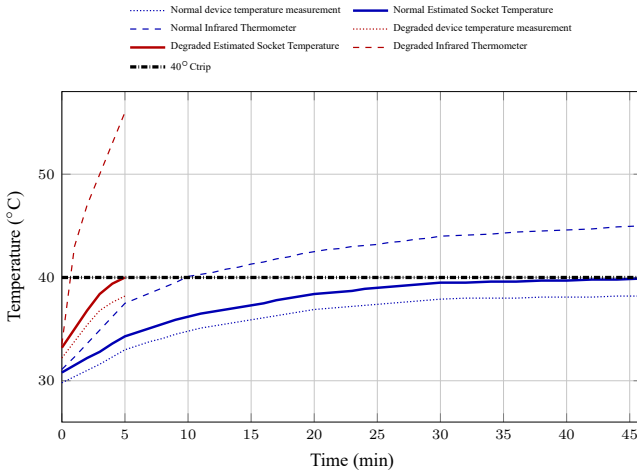


Fig. 6. Water heater plus four halogen lamps during high-current socket-heating protection validation.

Table V summarizes all socket-heating protection trials. At about 5A in the water-heater-only case, neither socket tripped: the corroded terminal ended hotter than the normal terminal (38.5°C versus 33.8°C), but the estimated socket temperature remained below the 40.0°C relay criterion. At about 9A, the corroded socket reached the 38.2°C comparison point in 48min, compared with 105min for the normal socket. At about 11A, the corroded socket reached the same point in 7min, compared with 52min for the normal socket. In the severe 13A corroded case, the relay tripped after 5min and the reference terminal

reached 56.0°C. These results support the premise that socket heating cannot be inferred from RMS current alone.

The difference between device temperature, estimated socket temperature, and IR terminal temperature is important. The prototype senses the contact hotspot indirectly through an external thermistor and a conservative relay-facing estimate. This approach provides socket-level action before contact heating progresses further while preserving non-trip behavior in low-current cases where the estimate remains below threshold.

TABLE V
Socket-Heating Protection Summary (WH: Water Heater)

Load case	Socket	Mean A	IR terminal °C	Relay
WH	Normal	5.17	33.8	No
WH	Corroded	4.85	38.5	No
WH + 1 lamp	Normal	7.11	40.3	No
WH + 1 lamp	Corroded	7.10	42.4	Yes
WH + 2 lamps	Normal	9.06	43.2	Yes
WH + 2 lamps	Corroded	9.04	44.1	Yes
WH + 3 lamps	Normal	11.07	43.9	Yes
WH + 3 lamps	Corroded	11.14	55.0	Yes
WH + 4 lamps	Normal	13.02	45.0	Yes
WH + 4 lamps	Corroded	13.08	56.0	Yes

D. Series Arc and TinyML Results

The cleaned arc-training dataset contained 49,296 rows with 283 arc markers, confirming that frame-level arc evidence was sparse and imbalanced. The final held-out Random Forest test split contained 10,268 feature rows, including 90 positive rows. Frame-level results were 10,173 true negatives, 5 false positives, 25 false negatives, and 65 true positives, corresponding to 99.7078% accuracy, 92.8571% precision, 72.2222% recall, and 81.25% F1-score. These row-level metrics show raw classifier behavior before a complete relay trial is interpreted.

Event-level metrics provide a stricter view of arc sensitivity over grouped episodes. The held-out event report contained 25 true arc events, 21 detected events, 4 missed events, 31 predicted events, 1 false-alarm event, 84.0% event recall, 96.7742% event precision, and 89.9358% event F1-score. These metrics show that some induced arc episodes were missed even though false event alarms remained low. They provide the raw event-sensitivity view that complements the relay-facing trial metrics.

The final relay-facing trial confusion matrix is shown in Table VI. Across 200 balanced normal and arc trials, the deployed model and firmware persistence logic detected 96 of 100 arc trials and produced one false trip in 100 normal trials. The corresponding performance metrics are shown in Table VII. Overall trial-level accuracy was 97.50%, precision was 98.97%, recall was 96.00%, and F1-score was 97.46%. Trial-level metrics are higher than frame-level and event-level metrics because they include the complete firmware behavior: repeated arc-like frames accumulate

toward a relay decision, while isolated normal outputs decay.

TABLE VI
Series Arc-Fault Trial Confusion Matrix

Trial group	TP	FP	FN	TN
Recognized loads	49	0	1	50
Unrecognized loads	47	1	3	49
All tested loads	96	1	4	99

TABLE VII
Series Arc-Fault Performance Metrics

Evaluation level	Accuracy	Precision	Recall	F1-score
Trial, recognized	99.00	100.00	98.00	98.99
Trial, unrecognized	96.00	97.92	94.00	95.92
Trial, all loads	97.50	98.97	96.00	97.46
Held-out frame	99.71	92.86	72.22	81.25
Held-out event	—	96.77	84.00	89.94

The residual trial-level errors were concentrated in brushed or universal-motor loads. The impact drill contributed the single normal false trip, while several brushed-load cases contributed missed arc trials. This behavior is consistent with load-dependent arc-fault literature: healthy motor commutation, startup, and nonlinear current features can overlap with arc indicators. The result supports load-aware inference within the tested socket-level classifier and its evaluated load-family boundaries.

The trial groups represent the collected load families. Water heater, halogen lamp, electric kettle, and heat gun cases were treated as resistive; the 40W stand fan represented the inductive family; and the handheld vacuum, Electrolux vacuum, angle grinder, bench cutter, and impact drill represented brushed or universal-motor loads. The dataset included difficult motor behavior while leaving broader household electronic-load coverage for future work.

E. Integrated Protection Assessment

The integrated assessment is that the three protection paths complement one another. Overload protection addressed persistent current stress without tripping every transient. Socket-heating protection responded to degraded-contact thermal behavior that current alone could not identify. Series arc protection used waveform features and context to address a load-limited fault that may remain below conventional overcurrent operation. These results indicate the value of separating evidence by fire-inducing mechanism while combining the outputs in one local relay-protection pipeline.

This assessment is bounded by the laboratory scope. The prototype demonstrated controlled feasibility under selected loads, induced faults, and bench conditions. Certification, field readiness, and full residential nuisance-immunity evaluation require separate future work.

F. Scope and Limitations

The study was conducted under controlled laboratory conditions. Long-term deployment in occupied homes, aged installations, large appliance populations, full branch-circuit diversity, and naturally occurring faults were outside the evaluation scope. The load set and fault conditions were limited by safe bench testing, especially for induced series arcs.

The arc model may produce nuisance detections or missed detections on unseen loads. Although the firmware context schema retained six possible family fields, the collected and tested load signatures were limited to resistive, inductive, and brushed or universal-motor loads. Sensitive SMPS loads, phase-angle-controlled loads, and mixed electronic loads were outside comprehensive induced-fault coverage. The reported false positive and false negative cases show that the dataset remains incomplete, especially for brushed and nonlinear loads. Reporting both event-level and trial-level metrics gives a more balanced view of raw sensitivity and relay-facing behavior.

The thermal sensing method is indirect. The thermistor was located outside the hidden contact spot, and the reference terminal reading often exceeded the external device temperature and estimated socket temperature. The 40.0°C threshold is a prototype control threshold for conservative relay action, not a universal socket safety limit or direct hotspot measurement.

Wi-Fi and Firebase supported monitoring, logging, labeling, and operator interaction. Local relay protection was independent of cloud connectivity, although remote visualization and history features were reduced when the network was unavailable. The device provides socket-level supervision only, without whole-house fault localization.

Certification-oriented testing was outside the present work. Creepage, clearance, enclosure robustness, relay thermal rating, endurance, electromagnetic compatibility, abnormal operation, and regulatory safety tests remain future work. The implementation should be evaluated as complementary socket-level supervision alongside breakers, AFCIs, GFCIs, surge protection, code-compliant wiring, and professional electrical assessment.

V. CONCLUSION AND RECOMMENDATIONS

The TinyML Smart Plug demonstrated controlled laboratory feasibility as a hybrid socket-level protection implementation for selected progressive fire-inducing electrical faults. Its ECE contribution is the integration of calibrated sensing, deterministic protection logic, load-aware embedded TinyML, relay latching, and monitoring support in one local protection pipeline. The work also shows why the evidence must be separated by mechanism: overload requires current persistence, degraded socket-contact heating requires conservative thermal estimation, and series arcs require load-aware waveform inference.

Within the tested scope, sensor errors were acceptable for the implemented thresholds, the sustained overload

function tripped the 17.30A case after 61.3s, degraded-contact tests showed faster and hotter socket behavior than normal contacts, and the deployed arc pipeline achieved 97.50% trial-level accuracy with missed-event and false-event cases retained in the analysis. These findings support feasibility for the proposed complementary protection architecture.

Future work should expand the normal, nuisance, and arc-fault dataset across more residential loads, plug conditions, contact gaps, and wiring arrangements. Longer semi-field testing should support later field-readiness evaluation. Thermal sensing should be improved through better placement, multiple sensors, or stronger estimation models. Firmware and hardware should be optimized for longer sampling, logging reliability, and memory margin. Enclosure, insulation, relay rating, creepage, clearance, strain relief, and thermal safety should be redesigned toward certification-oriented testing. Arc-event labeling should also be improved so future TinyML models can be trained and evaluated with stronger event-level evidence.

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